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Electronic device including cable connector

Abstract

An electronic device is disclosed, including: a plurality of electrical components, a substrate including a circuit electrically connected to the electrical component, wherein a through hole is formed penetrating the substrate in a thickness direction from an upper surface of the substrate, a cable, a head part coupled to an end portion of the cable, and a socket part including a contact pin of which at least a portion thereof is disposed inside the through hole and extends laterally with respect to a lower surface of the substrate, so as to be electrically connected to the substrate, and coupled to the head part in a direction of the upper surface of the substrate.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7320605	12/2007	Chang	439/74	H01R 12/52
8398425	12/2012	Suzuki	439/374	H01R 13/631
9124011	12/2014	Miyazaki	N/A	H01R 13/6271
9307644	12/2015	Kim	N/A	H01R 13/629
9344539	12/2015	Wang	N/A	G06F 1/1626
9484648	12/2015	Takenaga	N/A	H01R 12/716
10148031	12/2017	Tan	N/A	H01R 13/514
10389007	12/2018	Choi	N/A	H01Q 21/08
10726258	12/2019	Park	N/A	H04N 23/51
11398697	12/2021	Komoto	N/A	H01R 43/205
11799192	12/2022	Baek	N/A	H04B 5/24
2004/0157478	12/2003	Midorikawa	439/74	H01R 12/73
2013/0171876	12/2012	Funahashi	439/620.03	H01R 24/50
2014/0171159	12/2013	Endo	343/904	H01Q 1/243
2014/0232607	12/2013	Lee	343/841	H04B 15/04
2015/0323755	12/2014	Kwon	359/823	G02B 7/023
2017/0171440	12/2016	Park	N/A	G02B 7/08
2019/0012544	12/2018	Park	N/A	H04N 25/70
2019/0208045	12/2018	Xie	N/A	H05K 1/189
2019/0312334	12/2018	Shin	N/A	H04M 1/0249
2020/0288004	12/2019	Ren	N/A	G06F 3/041
2021/0006648	12/2020	Han	N/A	H01R 12/716
2021/0120113	12/2020	Shim	N/A	H04M 1/026
2021/0243291	12/2020	Lim et al.	N/A	N/A
2022/0201860	12/2021	Kim	N/A	H05K 1/181
2024/0089367	12/2023	Shim	N/A	H01Q 1/24

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
------------	------------------	---------	-----

210225475	12/2019	CN	N/A
11-144823	12/1998	JP	N/A
2002-42985	12/2001	JP	N/A
3396343	12/2002	JP	N/A
2006-261543	12/2005	JP	N/A
2013-106195	12/2012	JP	N/A
2014-120076	12/2013	JP	N/A
5790245	12/2014	JP	N/A
2020-191185	12/2019	JP	N/A
10-2008-0036813	12/2007	KR	N/A
10-2013-0012915	12/2012	KR	N/A
10-2016-0076733	12/2015	KR	N/A
10-2019-0103814	12/2018	KR	N/A
10-2021-0099264	12/2020	KR	N/A

OTHER PUBLICATIONS

International Search Report dated Mar. 2, 2023. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This is a continuation application that is based on and claims priority under 35 U.S.C. § 120 to PCT International Application No. PCT/KR2022/018253 which was filed on Nov. 18, 2022, and claims priority to Korean Patent Application No. 10-2021-0159468, filed on Nov. 18, 2021, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein their entirety.

TECHNICAL FIELD

(1) Certain embodiments disclosed in this document relate to an electronic device, and more particularly, to an electronic device including an internal cable connector.

DESCRIPTION OF RELATED ART

(2) Electronic devices include circuit boards. Circuit boards provide space and circuitry for disposing electrical connections between electronic components such as integrated circuits, passive elements, sensors, and cables for connection. The electrical components may be disposed directly on a surface of the circuit board, or be electrically connected to the circuit board through the cable.

(3) A cable maybe utilized for transmitting a signal or supplying power, such as a coaxial cable, a ribbon cable, or a flexible cable. The cable may be positioned within the electronic device. The cable may include a male connector on an end portion thereof, and the circuit board may include a female connector to which a male connector is coupled.

(4) The male connector may be disposed within a header that includes a stiffener providing rigidity for insertion into the female connector by an operator, or a robot during assembly of the electronic device. The female connector may be disposed in a socket to which the header is coupled, and the socket may be disposed on the circuit board via a dip pin, soldering, or surface mount technology (SMT).

SUMMARY

(5) When the female connector is disposed on a surface of the substrate in a direction so as to be coupled to the male connector, the socket and the header may protrude on the surface of the substrate. Thus, an excessively high height may be required for disposition of a cable connector, and adversely affect disposition of other components within the electronic device, and increase an overall thickness of the electronic device. Further, an increase in disposition height of the cable connector may increase a gap with a shielding material providing electromagnetic interference (EMI) shielding, thereby rendering EMI shielding incomplete.

(6) Certain embodiments disclosed in this document may provide an electronic device including a cable connector having a reduced disposition height.

(7) According to certain embodiments of the disclosure, an electronic device including a plurality of electrical components may include a substrate including a circuit electrically connected to the electrical component, wherein a through hole is formed penetrating the substrate in a thickness direction from an upper surface of the substrate, a cable, a head part coupled to an end portion of the cable, and a socket part including a contact pin of which at least a portion thereof is disposed inside the through hole and extends laterally with respect to a lower surface of the substrate, so as to be electrically connected to the substrate, and coupled to the head part in a direction of the upper surface of the substrate.

(8) In some embodiments, the electronic device may further include

(9) In some embodiments, the substrate may include an anchor fixed to the substrate by penetrating the substrate in a thickness direction, and configured to fix the support member to a lower surface thereof. In some embodiments, the electronic device may further include an auxiliary substrate positioned, in an area including an area in which the through hole is formed, on the lower surface of the substrate. In some embodiments, the substrate may include a first contact pad formed on a lower surface of the substrate and bonded to the contact pin of the socket part; and a plurality of second contact pads formed on the lower surface of the substrate and electrically connected to the circuit of the substrate, wherein the auxiliary substrate may include a bypass wiring coupled to the second contact pads to interconnect at least two of the second contact pads. In other embodiments, the auxiliary substrate may include a socket connection wiring bonded to the substrate and the contact pin of the socket part.

(10) The contact pin may be electrically connected to the substrate through the socket connection wiring of the auxiliary substrate. In some embodiments, a thickness of the auxiliary substrate may be thinner than that of the substrate. In other embodiments, the auxiliary substrate may include a flexible printed circuit board.

(11) In some embodiments, the electronic device may further include a second substrate positioned in a lower portion of the substrate and in which at least some of the electrical components are disposed. In some embodiments, the substrate may include a first contact pad formed on a lower surface of the substrate and bonded to the contact pin of the socket part; a plurality of second contact pads formed on the lower surface of the substrate and electrically connected to the circuit of the substrate, wherein the second substrate may include a bypass wiring coupled to the second contact pads to interconnect at least two of the second contact pads. In other embodiments, the second substrate may include a socket connection wiring bonded to the substrate and the contact pin of the socket part, and the contact pin may be electrically connected to the substrate through the socket connection wiring of the second substrate. In some embodiments, the electronic device may further include an interposer substrate interposed between the substrate and the second substrate, wherein the interposer substrate may include a via configured to electrically connect the substrate and the second substrate.

(12) In some embodiments, the substrate may include a first contact pad formed on a lower surface thereof and bonded to the contact pin of the socket part; and a plurality of second contact pads formed on the lower surface thereof and electrically connected to the circuit thereof, wherein the

interposer substrate may include a bypass wiring coupled to the second contact pads to interconnect at least two of the second contact pads. In some embodiments, the interposer substrate may include a socket connection wiring bonded to the substrate and the contact pin of the socket part, and the contact pin may be electrically connected to the substrate through the socket connection wiring of the interposer substrate.

(13) In some embodiments, the electronic device may further include a shielding member configured to cover an upper surface and a side surface of the head part and including a conductive material. In some embodiments, the shielding member may be formed by cutting the conductive material on the thin plate into a shape including a central portion and a plurality of side shielding parts extended radially from the central portion and by bending the side shielding part to cover the side surface of the head part. In some embodiments, the shielding member may be formed by cutting the conductive material on a thin plate into a T-shape or a plus (+) shape.

(14) According to certain embodiments disclosed in this document, because a socket is disposed at least partly in a through hole formed in the substrate, a height for disposing a cable connector on a substrate surface may be reduced. Thus, an electronic device equipped as such may be designed so as to have a reduced thickness, and improved EMI shielding capability may be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In connection with the description of the drawings, the same or similar reference numerals may be used for the same or similar components.

(2) FIG. 1 is a block diagram illustrating an example electronic device in a network environment according to certain embodiments;

(3) FIG. 2A is a perspective view illustrating an example front surface of an electronic device according to certain embodiments of the disclosure;

(4) FIG. 2B is a perspective view illustrating a rear surface of an example electronic device according to certain embodiments of the disclosure;

(5) FIG. 3 is an internal perspective view illustrating an example electronic device according to certain embodiments of the disclosure;

(6) FIG. 4A is an exploded perspective view illustrating an example electronic device according to certain embodiments of the disclosure;

(7) FIG. 4B is a cross-sectional view illustrating an example electronic device according to certain embodiments of the disclosure;

(8) FIG. 5A is a side view illustrating a head part and a socket part of an example electronic device according to embodiments of the disclosure;

(9) FIG. 5B is a side cross-sectional view illustrating an example socket part and a head part according to embodiments of the disclosure;

(10) FIG. 5C is a plan view and a side view illustrating an example socket part according to embodiments of the disclosure;

(11) FIG. 5D is a side view illustrating coupling of an example head part and a socket part of an electronic device according to embodiments of the disclosure;

(12) FIG. 6A is a cross-sectional view illustrating an example support member according to some embodiments of the disclosure;

(13) FIG. 6B is a cross-sectional view illustrating an example support member according to other embodiments of the disclosure;

(14) FIG. 7A is a cross-sectional view illustrating a socket part of an example electronic device according to other embodiments of the disclosure;

(15) FIG. 7B is a plan view illustrating an example auxiliary substrate of an electronic device

according to other embodiments of the disclosure;

(16) FIG. 7C is a plan view illustrating a contact pad of an example electronic device according to other embodiments of the disclosure;

(17) FIG. 7D is a side view illustrating a coupling relationship between a substrate, a socket part, and an auxiliary substrate of an example electronic device according to another embodiment of the disclosure;

(18) FIG. 8 is a cross-sectional view illustrating a socket part and an auxiliary substrate of an example electronic device according to another embodiment of the disclosure;

(19) FIG. 9 is a cross-sectional view illustrating a socket part and a second substrate of an example electronic device according to another embodiment of the disclosure;

(20) FIG. 10A is a plan view illustrating a shielding member of an example electronic device according to some embodiments of the disclosure; and

(21) FIG. 10B is a side view illustrating a shielding member of an example electronic device according to some embodiments of the disclosure.

DETAILED DESCRIPTION

(22) FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to certain embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network).

According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**). **11**

(23) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation.

According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(24) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor

121 while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(25) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(26) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(27) The input **1** module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input **1** module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(28) The sound output **1** module **155** may output sound signals to the outside of the electronic device **101**. The sound output **1** module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(29) The display **1** module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display **1** module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display **1** module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(30) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input **1** module **150**, or output the sound via the sound output **1** module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

(31) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(32) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(33) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

(34) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(35) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(36) The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(37) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(38) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(39) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band)

to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(40) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element implemented using a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(41) According to certain embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(42) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(43) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example.

(44) FIG. 2A is a perspective view illustrating a front surface of an electronic device according to

certain embodiments of the disclosure.

(45) FIG. 2B is a perspective view illustrating a rear surface of an electronic device according to certain embodiments of the disclosure.

(46) With reference to FIGS. 2A and 2B, an electronic device **200** according to an embodiment may include a housing **210** including a first surface (or front surface) **210A**, a second surface (or rear surface) **210B**, and a side surface **210C** enclosing a space between the first surface **210A** and the second surface **210B**. In another embodiment (not illustrated), the housing may refer to a structure forming a portion of the first surface **210A**, the second surface **210B**, and the side surface **210C** of FIG. 2A. According to an embodiment, the first surface **210A** may be formed by a front plate **202** (e.g., a polymer plate or a glass plate including various coating layers) that is at least partially substantially transparent. The second surface **210B** may be formed by a substantially opaque rear plate **211**. The rear plate **211** may be formed by, for example, coated or colored glass, ceramic, polymer, metal (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of at least two of the above materials. The side surface **210C** may be coupled to the front plate **202** and the rear plate **211** and be formed by a side bezel structure (or “side member”) **218** including a metal and/or a polymer. In some embodiments, the rear plate **211** and the side bezel structure **218** may be integrally formed and include the same material (e.g., a metal material such as aluminum).

(47) In the illustrated embodiment, the front plate **202** may include two first areas **210D** extended seamlessly by bending from the first surface **210A** toward the rear plate **211** in both ends of a long edge thereof. In the illustrated embodiment (see FIG. 2B), the rear plate **211** may include two second areas **210E** extended seamlessly by bending from the second surface **210B** toward the front plate **202** at both ends of the long edge. In some embodiments, the front plate **202** (or the rear plate **211**) may include one of the first areas **210D** (or the second areas **210E**). In another embodiment, the front plate **202** (or the rear plate **211**) may not include some of the first areas **210D** or the second areas **210E**. In the above embodiments, when viewed from the side surface of the electronic device **200**, the side bezel structure **218** may have a first thickness (or width) at a side surface that does not include the first areas **210D** or the second areas **210E** and have a second thickness thinner than the first thickness at the side surface including the first areas **210D** or the second areas **210E**.

(48) According to an embodiment, the electronic device **200** may include at least one of a display **201**, audio modules **203**, **207**, and **214**, sensor modules **204**, **216**, and **219**, camera modules **205**, **212**, and **213**, a key input device **217**, a light emitting element **206**, or connector holes **208** and **209**. In some embodiments, the electronic device **200** may omit at least one (e.g., the key input device **217** or the light emitting element **206**) of the components or may additionally include other components.

(49) The display **201** may be visually exposed, for example, through a substantial portion of the front plate **202**. In some embodiments, at least a portion of the display **201** may be visually exposed through the front plate **202** forming the first surface **210A** and the first areas **210D** of the side surface **210C**. In some embodiments, an edge of the display **201** may be formed to be substantially the same as an adjacent outer shape of the front plate **202**. In another embodiment (not illustrated), in order to expand an area in which the display **201** is exposed, the distance between an outer edge of the display **201** and an outer edge of the front plate **202** may be substantially equally formed.

(50) In another embodiment (not illustrated), a recess or opening may be formed in a portion of a screen display area of the display **201**, and the electronic device **200** may include at least one of the audio module **214**, the sensor module **204**, the camera module **205**, or the light emitting element **206** aligned with the recess or the opening. In another embodiment (not illustrated), at a rear surface of the screen display area of the display **201**, at least one of the audio module **214**, the sensor module **204**, the camera module **205**, a fingerprint sensor **216**, or the light emitting element **206** may be formed. In another embodiment (not illustrated), the display **201** may be coupled to or be disposed to adjacent to a touch sensing circuit, a pressure sensor capable of measuring the intensity (pressure) of a touch, and/or a digitizer that detects a magnetic field type stylus pen. In

some embodiments, at least a portion of the sensor modules **204** and **219** and/or at least a portion of the key input device **217** may be disposed in the first areas **210D** and/or the second areas **210E**.

(51) The audio modules **203**, **207**, and **214** may include a microphone hole and speaker holes. A microphone for acquiring an external sound may be disposed inside the microphone hole, and in some embodiments, a plurality of microphones may be disposed to detect a direction of a sound. The speaker holes **207** and **214** may include an external speaker hole and a receiver hole for a call. In some embodiments, the speaker holes and the microphone hole may be implemented into a single hole or a speaker (e.g., piezo speaker) may be formed without the speaker holes.

(52) The sensor modules **204**, **216**, and **219** may generate electrical signals or data values corresponding to an internal operating state or an external environmental state of the electronic device **200**. The sensor modules **204**, **216**, and **219** may include, for example, a first sensor module **204** (e.g., proximity sensor) disposed at the first surface **210A** of the housing **210**, a second sensor module (not illustrated) (e.g., fingerprint sensor), a third sensor module **219** (e.g., HRM sensor), and/or a fourth sensor module (e.g., fingerprint sensor) **216** disposed at the second surface **210B** of the housing **210**. The fingerprint sensor may be disposed at the first surface **210A** (e.g., the display **201**) as well as the second surface **210B** of the housing **210**. The electronic device **200** may further include a sensor module (not illustrated), for example, at least one of a gesture sensor, a gyro sensor, a barometric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor **204**.

(53) The camera modules **205**, **212**, and **213** may include a first camera device **205** disposed at the first surface **210A** of the electronic device **200**, a second camera device **212** disposed at the second surface **210B** of the electronic device **200**, and/or a flash **213**. The camera devices **205** and **212** may include one or more lenses, an image sensor, and/or an image signal processor. The flash **213** may include, for example, a light emitting diode or a xenon lamp. In some embodiments, two or more lenses (infrared cameras, wide-angle and telephoto lenses) and image sensors may be disposed on one surface of the electronic device **200**.

(54) The key input device **217** may be disposed at the side surface **210C** of the housing **210**. In other embodiments, the electronic device **200** may not include some or all of the above-mentioned key input devices **217**, and a non-included key input device **217** may be implemented in another form, such as a soft key, on the display **201**. In some embodiments, the key input device may include a sensor module **216** disposed at the second surface **210B** of the housing **210**.

(55) The light emitting element **206** may be disposed, for example, on the first surface **210A** of the housing **210**. The light emitting element **206** may provide, for example, state information of the electronic device **200** in the form of light. In another embodiment, the light emitting element **206** may provide, for example, a light source interworked with an operation of the camera module **205**. The light emitting element **206** may include, for example, LEDs, IR LEDs, and xenon lamps.

(56) The connector holes **208** and **309** may include a first connector hole **208** capable of receiving a connector (e.g., USB connector) for transmitting and receiving power and/or data to and from an external electronic device and/or and a second connector hole (e.g., earphone jack) **209** capable of receiving a connector for transmitting and receiving audio signals to and from an external electronic device.

(57) FIG. 3 is an internal perspective view illustrating an electronic device according to certain embodiments of the disclosure.

(58) With reference to FIG. 3, an electronic device **300** may include a side bezel structure **310** (e.g., housing), a support member **311** (e.g., bracket), a first printed circuit board **320**, a second printed circuit board **340**, and a battery **350**. In some embodiments, the electronic device **300** may omit at least one of the components or may additionally include other components. At least one of the components of the electronic device **300** may be the same as or similar to at least one of the components of the electronic device **200** of FIG. 2A or 2B, and a duplicate description thereof will

be omitted below.

(59) The first support member **311** may be disposed inside the electronic device **300** so as to be connected to the side bezel structure **310** (e.g., housing), or may be integrally formed with the side bezel structure **310**. The first support member **311** may be made of, for example, a metal material and/or a non-metal (e.g., polymer) material. The first support member **311** may include a display (not illustrated) coupled to one surface thereof and first, and second printed circuit boards **320** and **340** coupled to the other surface thereof. A processor, a memory, a sensor, a camera, and/or a female connector **321** may be mounted on the first printed circuit board **320**. The processor may include, for example, one or more of a central processor, an application processor, a graphic processor, an image signal processor, a sensor hub processor, or a communication processor. In some embodiments, the first printed circuit board may be a laminated substrate on which a plurality of PCBs are stacked.

(60) The female connector **321** electrically connect a cable **322** to the electrical component of the electronic device via coupling to a male connector **323** positioned in an end portion of the cable **322**. The term “male” may refer to a connector positioned in an end portion of a cable, and the term “female” may refer to a connector positioned at an object, to which the cable is electrically connected. The female connector and the male connector may be coupled via an interference fit by a frictional force, or snap fit coupling using an elastic lug. In some embodiments, the female connector may include a receiving portion in which a conductive contact portion is formed, and the male connector may have a terminal insertable into the female connector so as to form an electrical contact in the conductive contact portion, but the disclosure is not limited thereto and a configuration opposite to the above-described configuration is also possible.

(61) The cable **322** may be a member that interconnects electrical components such as, for example, the first and second printed circuit boards **320** and **340**, an antenna (not illustrated), a camera, and/or a display panel (e.g., the display **201** of FIG. 2A) of the electronic device. In some embodiments, the cable may include a ribbon cable, a coaxial cable, a flexible flat cable (FFC), or a flexible printed cable (FPC). A male connector may be disposed in an end portion of the cable.

(62) The second printed circuit board **340** may include another electrical component of the electronic device, for example, a power management IC (PMIC) external interface, an interface (e.g., USB interface) for transmitting and receiving power and/or data to and from an external electronic device) and/or an interface (e.g., 3.5 mm earphone jack) for transmitting and receiving analog audio signals to and from an external electronic device. The second printed circuit board may include a cable **322** for connection with the first printed circuit board.

(63) The battery **350** may supply power to at least one component of the electronic device **300**, and may include, for example, a non-rechargeable primary battery, a rechargeable secondary battery, and/or a fuel cell. At least a portion of the battery **350** may be, for example, disposed on substantially a same plane as that of the first and second printed circuit boards **320** and **340**. The battery **350** may be integrally disposed inside the electronic device **300** or may be disposed detachably from the electronic device **300**.

(64) The electronic device **300** according to certain embodiments may include an electronic device such as a bar type, a foldable type, a rollable type, a sliding type, a wearable type, and a tablet PC and/or a notebook PC. The electronic device **300** according to certain embodiments of the disclosure is not limited to the above-described example, and may include various other electronic devices.

(65) FIG. 4A is an exploded perspective view illustrating an electronic device according to certain embodiments of the disclosure.

(66) FIG. 4B is a cross-sectional view illustrating an electronic device according to certain embodiments of the disclosure.

(67) FIGS. 4A and 4B are cross-sectional views taken along line A-A of FIG. 3.

(68) With reference to FIGS. 4A and 4B, an electronic device **400** may include a display **401** (e.g.,

the display **201** of FIG. 2A), a front bracket **402**, a first substrate **410** (e.g., the first printed circuit board **320** or the second printed circuit board **340** of FIG. 3), an electrical component **403** (e.g., camera), a cable **420** (e.g., the cable **322** of FIG. 3), a head part **430**, a socket part **440**, a through hole **411**, and a rear bracket **404** (e.g., the support member **311** of FIG. 3).

(69) The electrical component **403** is an active or passive component electrically operated to perform a function of an electronic device, and may include a processor, a sensor, a camera, a display driver integrated circuit (DDIC), a power management integrated circuit (PMIC), and a battery. The electrical component may be disposed on the first substrate **410** or electrically connected to the first substrate **410** by the cable **420**. The first substrate **410** may be a printed circuit board. In some embodiments, the printed circuit board may include a laminated substrate including a plurality of substrates and/or an interposer substrate.

(70) The head part **430** is formed in an end portion of the cable **420**, and may be coupled to an object to which the cable **420** is to be connected, such as, for example, the socket part **440** (to be described later) formed on the first substrate **410** so as to form an electrical connection. A male connector **432** may be positioned in the head part **430**. The socket part **440** is disposed at the first substrate **410** to which the cable **420** is connected and is electrically connected to the first substrate **410**, and is coupled to the head part **430** so as to electrically connect the first substrate **410** with the cable **420**, and to fix the head part **430**. The through hole **411** may include a portion of the first substrate **410** that is formed so as to penetrate the first substrate **410** in a direction from an upper surface (i.e., meaning a surface of a portion of the first substrate **410** to which the head part **430** of the cable **420** is coupled and located, as described later below) to a lower surface of the first substrate **410** and which includes an internal space. At least a portion of the socket part **440** may be inserted and disposed in the through hole **411** formed in the first substrate **410**. When the first substrate **410** is a laminated substrate, the through hole **411** may extend downward so as to penetrate the plurality of substrates and/or the interposer substrates included in the laminated substrate, and heights of the head part **430** and the socket part **440** may extend so as to correspond to an extended depth of the through hole **411**. A detailed configuration of the head part **430** and the socket part **440** will be described later with reference to the drawings.

(71) FIG. 5A is a side view illustrating ahead part **430** and a socket part **440** of an electronic device, according to embodiments of the disclosure.

(72) FIG. 5B is a side cross-sectional view illustrating a socket part **440** and a head part **430** according to embodiments of the disclosure.

(73) FIG. 5C is a plan view and a side view illustrating a socket part **440** according to embodiments of the disclosure.

(74) FIG. 5D is a side view illustrating coupling of a head part **430** and a socket part **440** of an electronic device according to embodiments of the disclosure.

(75) With reference to FIGS. 5A and 5B, the first substrate **410** of the electronic device may include a through hole **411**, a socket part **440** that is disposed inside the through hole **411**, and the head part **430** positioned in an end portion of the cable **420** so as to be coupled to the socket part **440**.

(76) The head part **430** may be positioned in an end portion of the cable **420** so as to be coupled to the socket part **440**, thereby forming an electrical connection. The head part **430** may include a male connector **432** for forming an electrical contact with a female connector **441** (to be described later) of the socket part **440**, and a head stiffener **431** for protecting the head part **430** from any stress applied to the head part **430** during assembly the electronic device. The male connector **432** may include a plurality of male connector pins **432a**. The male connector pins **432a** may include a conductive material, and be arranged in one or more rows. An end portion of the male connector pin **432a** may be hemmed or bent so as to form a curved surface, thereby stably establishing electrical contact with the female connector **441**. In some embodiments, the male connector pins **432a** may be arranged in two opposing rows. The male connector pins **432a** in rows facing one

another may be arranged so as to be symmetrical to each other. In other embodiments, the male connector pins **432a** may be arranged in rows of four or more.

(77) The through hole **411** may have a planar shape, corresponding to a planar shape of the socket part **440**, and penetrate the first substrate **410** in a thickness direction, and have an area in which the socket part **440** is positioned therein. In some embodiments, a contact pad **412** for electrically bonding to the socket part **440** may be formed in an area around the through hole **411** on a lower surface of the first substrate **410**. The contact pad **412** may include a conductive material including copper, gold, nickel, tin or an alloy thereof and plating.

(78) In some embodiments, a support member **450** and/or an auxiliary substrate **460** (e.g., an auxiliary substrate **460** of FIG. 7A) for supporting the socket part **440** may be positioned in a lower portion of the through hole **411**. Detailed implementations of the support member **450** and the auxiliary substrate **460** will be described later.

(79) With reference to FIG. 5C, the socket part **440** may be positioned inside the through hole **411**, electrically connected to the first substrate **410**, and coupled to the head part **430** so as to electrically connect the cable **420** and the first substrate **410**. The socket part **440** may include a female connector **441** for forming electrical contact with the male connector **432**, a contact pin **442** extending laterally with respect to a lower surface of the first substrate **410** from the female connector **441** and electrically connected to the first substrate **410**, and a socket housing **443** for providing an internal space for disposition of the female connector **441** and protecting and/or supporting the female connector **441**. The contact pin **442** may extend laterally from a lower portion of the socket part **440** so as to be bonded to the contact pad **412**, as formed on a lower surface of the first substrate **410**. For bonding the contact pin **442** and the contact pad **412**, soldering, spot welding, surface mount technology (SMT), or a technical implementation for forming an electrical bonding similar thereto may be used. With reference to FIG. 5B, the female connector **441** may include a metal material processed to stably form an electrical contact surface with the male connector **432** by pressing of the male connector **432**, via elastic deformation when coupled with the male connector **432**.

(80) Because the socket part **440** is positioned inside the through hole **411**, a disposition height of the socket part **440**, the head part **430** (as coupled to an upper portion of the socket part **440**), and the cable **420** may be lowered. Therefore, it may provide advantages enabling reduction of a thickness of the electronic device. Further, when a lateral force is applied to the cable **420**, a corresponding amount of lateral force applied to the socket part **440**, the head part **430**, and the first substrate **410** may be reduced. Thus, there is an advantage that the risk of damage is reduced. Further, because the socket part **440** is positioned inside the through hole **411**, more guidance is provided to the head part **430** when the through hole **411** is being coupled to the head part **430**. Further, because risk of damage to the head part **430** is reduced, a thickness of the stiffener **431** of the head part **430** may be reduced, thereby reducing an overall thickness. In some embodiments, the height of the socket part **440** is greater than a thickness of the first substrate **410** in an area in which the through hole **411** is formed. Thus, the socket part **440** may be stably coupled to the head part **430**.

(81) With reference to FIG. 5D, the head part **430** may be coupled to the socket part **440** by pressing against the socket part **440** from an upper part (i.e., in the z-axis direction of FIG. 5D) of the first substrate **410**. The head part **430** and the socket part **440** may be coupled via detachable coupling, such as by an interference fit, a clasp, a screw, or snap coupling.

(82) FIG. 6A is a cross-sectional view illustrating a support member **450** according to some embodiments of the disclosure.

(83) FIG. 6B is a cross-sectional view illustrating a support member **450** according to other embodiments of the disclosure.

(84) With reference to FIG. 6A, the support member **450** may be disposed in a bottommost portion of an area on a lower surface (e.g., assuming upwards in FIG. 6A is “higher” and downwards in

FIG. 6A is “lower”) of the first substrate **410** in which the socket part **440** is positioned. The support member **450** supports a bottommost portion of the socket part **440** to reduce the risk that the socket part **440** may be damaged or dropped, according to a pressure applied from the header part to the socket part **440**, in order to couple the header part to the socket part **440** upon assembling the electronic device. The support member **450** may include a material such as at least one of metal, ceramic, or synthetic resin. The support member **450** may include a metal material such as stainless steel. Because the metal material has high specific strength and low modulus of elasticity, and the metal material is conductive, the metal material may provide an electro-magnetic interference (EMI) shielding effect for the socket part **440** and grounding of the circuit of the electronic device.

(85) In some embodiments, the first substrate **410** may include a coupling pad **413** disposed in an area adjacent to the through hole **411** on a lower surface thereof (e.g., near an underside of the first substrate **410**, assuming downwards in FIG. 6A is “lower”), which may be coupled to the support member **450**. The coupling pad **413** of the support member **450** may include a conductive material including copper, gold, nickel, tin or an alloy thereof and plating. The support member **450** may be coupled to the coupling pad **413** by a method such as soldering, spot welding, or SMT. The support member **450** may be fixed to the lower surface of the first substrate **410** by being bonded to the coupling pad **413** to support the socket part **440**. In another embodiment, the support member **450** may be coupled to the first substrate **410** by a fastening means such as a screw or a snap rivet.

(86) In some embodiments, the first substrate **410** may include a plurality of ground wirings **415** for grounding electrical components of the electronic device, and the coupling pad **413** of the support member **450** may include a ground pad **412c** electrically connected to the ground wirings. The support member **450** including a metal material may be connected to the ground wiring of the first substrate **410** through the ground pad **412c**. Thus, the support member **450** may connect the plurality of ground wirings positioned around the through hole **411** to one another, thereby completing the ground circuit, and EMI noise signals may thus be sent through the ground. Thus, the support member **450** may have an EMI shielding effect.

(87) With reference to FIG. 6B, in another embodiment, the first substrate **410** may include an anchor **414**. The anchor **414** may penetrate the first substrate **410** in a thickness direction of the first substrate **410**, and in an area of the first substrate **410** adjacent to the through hole **411**. The anchor **414** may be coupled to the support member **450** to fix the support member **450** to the first substrate **410**. Because the anchor **414** penetrates the first substrate **410** to so as to fix the support member **450** with respect to the entire thickness of the first substrate **410**, a fixing force for the support member **450** may be improved. Further, the anchor **414** may electrically connect the support member **450** to a plurality of ground wirings **415** positioned at a lower surface, and an upper surface of the first substrate **410**, and a central portion in a thickness direction of the first substrate **410**.

(88) FIG. 7A is a cross-sectional view illustrating a socket part **440** of an electronic device according to other embodiments of the disclosure.

(89) FIG. 7B is a plan view illustrating an auxiliary substrate **460** of an electronic device according to other embodiments of the disclosure.

(90) FIG. 7C is a plan view illustrating a contact pad **412** of an electronic device according to other embodiments of the disclosure.

(91) FIG. 7D is a side view illustrating a coupling relationship between a first substrate **410**, a socket part **440**, and an auxiliary substrate **460** of an electronic device according to other embodiments of the disclosure.

(92) With reference to FIGS. 7A and 7B, the electronic device may include an auxiliary substrate **460** positioned in a bottommost portion of the first substrate **410** (i.e., with the same up-down reference perspective as FIGS. 6A-6B). The auxiliary substrate **460** may be positioned in an area including the through hole **411** and a peripheral portion of the through hole **411** on the lower

surface of the first substrate **410**. The auxiliary substrate **460** may include a bypass wiring **461**, and a bypass wiring contact pad **462** in which an electrical contact for connection of the bypass wiring **461** is formed may be formed on the auxiliary substrate **460**. The bypass wiring **461** may be a circuit connecting circuits of the first substrate **410** that are otherwise cut off by the through hole **411** of the first substrate **410**. The auxiliary substrate **460** may include a printed circuit board, a flexible printed circuit board (FPCB), an integrated circuit, a low temperature co-sintered ceramic (LTCC), or a similar substrate in which an electric circuit is formed. In some embodiments, the auxiliary substrate **460** may include a plurality of bypass wirings **461** oriented in different directions. The auxiliary substrate **460** may include a laminated substrate having a plurality of layers, and bypass wirings **461** oriented in different directions may be formed in each layer. In some embodiments, in order to save an internal space of the electronic device, the thickness of the auxiliary substrate **460** may be thinner than that of the first substrate **410**. Because a structure of the circuit (e.g., such as the bypass wiring **461** disposed at the auxiliary substrate **460**) is simple, the circuit may be implemented so as to have a desired thinness. For example, the auxiliary substrate **460** may include a flexible printed circuit board in which a bypass wiring **461** is formed by printing a conductor on a plastic plate, resulting in a desired thinness. Because the auxiliary substrate **460** includes a flexible printed circuit board, the bypass wiring **461** may be applied while reducing an increase in thickness due to the addition of the auxiliary substrate **460**. In another embodiment, the auxiliary substrate **460** may be produced using a residual substrate generated when the first substrate **410** is produced by cutting a substrate mother sheet. By producing the auxiliary substrate **460** using the residual substrate, waste may be reduced and material costs may be reduced.

(93) With reference to FIG. 7C, the contact pad **412** of the lower surface of the first substrate **410** may include a first contact pad **412a** bonded to the contact pin **442** of the socket part **440**, and second contact pads **412b** connected to the circuit of the first substrate **410**. The first contact pad **412a** may include an electrical contact point for connecting the cable **420** and the first substrate **410** through the socket part **440** and the head part **430**. The second contact pads **412b** may electrically connect to the bypass wiring **461** by bonding to the bypass wiring contact pad **462** when the auxiliary substrate **460** is coupled to the lower portion of the first substrate **410**. The circuit of the first substrate **410**, as cut off by the through hole **411**, may be electrically connected to the bypass wiring **461** through the second contact pads **412b** and the bypass wiring contact pad **462** at a point in the periphery of the through hole **411**, and the bypass wiring **461** may be reconnected to the first substrate **410** through the second contact pads **412b** and the bypass wiring contact pad **462** positioned at different points in the periphery of the through hole. Accordingly, the circuit of the first substrate **410** can bypass the disconnection caused by the through hole **411**. In some embodiments, the first contact pad **412a** may be formed in an area relatively close to the through hole **411** and the socket part **440** positioned within the through hole **411** among areas in which the contact pad **412** is formed, and the second contact pads **412b** may be positioned at a periphery of an area in which the first contact pad **412a** is formed.

(94) In case of forming the through hole **411** in the first substrate **410**, the circuit around the through hole **411** may bypass the through hole **411**. In order to dispose the circuit so as to bypass the through hole **411**, an area of an additional first substrate **410** may be utilized, which may interfere with downsizing of the first substrate **410**. The bypass wiring **461** of the auxiliary substrate **460** may reduce requisite utilization of an area of an additional first substrate **410** for bypassing the through hole **411**, by allowing the bypass through a lower portion of an area of the substrate in which the through hole **411** of the first substrate **410** is formed.

(95) With reference to FIG. 7D (a), a solder ball **416** may be utilized to bond with the first contact pad **412a**, the second contact pads **412b** of the first substrate **410** may be attached to the socket part **440** and the auxiliary substrate **460**, and the socket part **440** and the auxiliary substrate **460** may be seated on lower surface of the first substrate **410** in an inverted state, such that the lower surface faces upward. With reference to FIG. 7D (b), when in a state in which the socket part **440** and the

auxiliary substrate **460** are seated on the lower surface of the first substrate **410**, the solder ball **416** is heated and melted. Thus, the socket part **440** and the auxiliary substrate **460** may be bonded to the first contact pad **412a** and the second contact pads **412b**. An electric resistance, a laser, or an oven may be used as methods of heating the solder ball **416**. With reference to FIG. 7D (c), the first substrate **410** in which bonding is completed may be returned to an original position thereof, so that an upper surface thereof faces upwards again. By the above-described process, the socket part **440** and the auxiliary substrate **460** are bonded to the first substrate **410**; thus, the socket part **440** and the auxiliary substrate **460** may be simultaneously bonded to the first substrate **410** through one heating. During a bonding process of the socket part **440** and the auxiliary substrate **460**, various other electrical components disposed on the first substrate **410** are simultaneously bonded, thereby simplifying the process.

(96) FIG. **8** is a cross-sectional view illustrating a socket part **440** and an auxiliary substrate **460** of an electronic device according to another embodiment of the disclosure.

(97) With reference to FIG. **8**, the auxiliary substrate **460** may include a socket connection wiring **463** for electrically connecting the contact pin **442** of the socket part **440** to the first substrate **410**, and an auxiliary substrate contact pad **464** for bonding the socket connection wiring **463** and the contact pad **412** of the first substrate **410**. Accordingly, the socket part **440** may be connected to the first substrate **410** through the auxiliary substrate **460**. In an embodiment in which the socket part **440** is electrically connected to the first substrate **410** through the auxiliary substrate **460**, a comparatively larger portion of the socket part may be inserted into the through hole. Thus, the socket part **440** may be so disposed while reducing the degree of external protrusion thereof from the lower surface of the first substrate **410** of the through hole, as compared to an embodiment in which the socket part **440** is directly connected to the contact pad **412** as formed at the lower surface of the substrate. Further, because the socket part **440** is disposed at the auxiliary substrate **460**, a degree of freedom in disposing the first substrate **410** may be increased.

(98) FIG. **9** is a cross-sectional view illustrating a socket part **440** and a second substrate **470** of an electronic device according to another embodiment of the disclosure.

(99) With reference to FIG. **9**, the electronic device may include a second substrate **470** disposed in a lower portion of the first substrate **410**. The second substrate **470** may include a circuit for providing an additional disposition space for installation of electrical components such as a coprocessor, a storage device, an antenna, a PMIC, and/or a DDIC, and for electrically connecting the first substrate **410** with the above-described additional electrical components. The second substrate **470** may include bypass wiring **461** and/or socket connection wiring **463**. Regarding the bypass wiring **461** and the socket connection wiring **463**, the description of the auxiliary substrate **460** may be referred to as long as there is no contradiction.

(100) In some embodiments, an interposer substrate **480** may be disposed between the first substrate **410** and the second substrate **470**. The interposer substrate **480** may include a conductive via **481** penetrating the interposer substrate **480** in a thickness direction, so as to electrically connect the first substrate **410** and the second substrate **470** and/or a connection wiring **482** for electrically connecting an electrical component disposed at the first substrate **410** and an electrical component disposed at the second substrate **470**. The connection wiring **482** may include a first via **482a** electrically connected to the first substrate **410**, a second via **482c** electrically connected to the second substrate **470**, and horizontal wiring **482b** for connecting the first via **482a** and the second via **482c**. The interposer substrate **480** including the connection wiring **482** may be a multilayer substrate that includes a plurality of layers.

(101) The interposer substrate **480** may include a bypass wiring **461** and/or a socket connection wiring **463**. Regarding the bypass wiring **461** and the socket connection wiring **463**, the description of the auxiliary substrate **460** provided previously may be referred to, as long as there are no contradictions.

(102) The second substrate **470** may help miniaturization of the electronic device by reducing an

area of the first substrate **410**, and the interposer substrate **480** may increase the freedom with which components may be arranged on the first substrate **410** and the second substrate **470**. By thus locating the bypass wiring **461** and/or the socket connection wiring **463** at the second substrate **470** or the interposer substrate **480**, the bypass wiring **461** and/or the socket connection wiring **463** may be so disposed without incurring increases in overall thickness by the auxiliary substrate **460**.

(103) FIG. **10A** is a plan view illustrating a shielding member **490** of an electronic device according to some embodiments of the disclosure.

(104) FIG. **10B** is a side view illustrating a shielding member **490** of an electronic device according to some embodiments of the disclosure.

(105) With reference to FIGS. **10A** and **10B**, the shielding member **490** may include a plate-shaped member of a desired thickness covering an upper surface and a side surface of the head part **430** on a surface of the first substrate **410**, and/or an upper part of the first substrate **410** adjacent to the head part **430**. The shielding member **490** may be electrically connected to a ground wiring of the first substrate **410**.

(106) The shielding member **490** may be a plate-shaped member made having the desired thickness, formed of a conductive material that shields EMI generated in an area in which the first substrate **410** and the cable **420** are coupled, and/or EMI introduced from the external environment. The shielding member **490** may include a metal foil, a metal mesh, or a thin metal-coated synthetic resin plate. The metal foil or metal mesh may be freely formed from a general-purpose material. In some embodiments, the shielding member **490** may be formed by cutting from the above-described thin plate-shaped material into a shape, for example, a T-shape or a plus (+) shape having a plurality of side shielding parts **490b** radially extended from a central portion **490a**. The shielding member **490** may be bent such that the central portion **490a** covers an upper surface of the head part and the side shielding parts shield the head part and the plurality of side surfaces.

(107) When the mounting height of the head part **430** is sufficiently high, a gap between the side shielding parts **490b** of the shielding member **490** covering the side surface of the head part **430** may be widened, thereby increasing the risk of EMI noise leakage. According to certain embodiments of the disclosure, because the shielding member **490** is positioned inside the through hole **411** formed in the first substrate **410**, a height at which the head part **430** and the socket part **440** protrude from the first substrate **410** may be reduced. Accordingly, the risk of leakage of EMI noise from the shielding member **490** may be reduced, and a shielding performance of the shielding member **490** may thus be improved.

(108) Embodiments disclosed in this document disclosed in this specification and drawings merely present specific examples to easily describe the technical content according to the embodiments disclosed in this document and to help the understanding of the embodiments disclosed in this document, and are not intended to limit the embodiments disclosed in this document. Accordingly, in addition to the embodiments disclosed herein, all changes or modifications derived from the technical ideas of certain embodiments disclosed in this document should be construed as being included in the embodiments disclosed in this document.

Claims

1. An electronic device including a plurality of electrical components, the electronic device comprising: a first substrate including a circuit electrically connected to an electrical component from among the plurality of electrical components, wherein a through hole is formed penetrating the first substrate in a thickness direction from an upper surface of the first substrate; a cable; a head part coupled to an end portion of the cable; a socket part including a contact pin of which at least a portion is disposed inside the through hole and extending with respect to a lower surface of the first substrate, the socket part being electrically connected to the first substrate, and coupled to the head part in a direction of the upper surface of the first substrate; and a second substrate

positioned on below an area of the lower surface of the first substrate in which the through hole is formed, wherein the first substrate comprises: a plurality of contact pads formed at the lower surface of the first substrate, and electrically connected to the circuit of the first substrate, wherein the second substrate includes wiring coupled to at least two of the plurality of contact pads so as to interconnect the at least two of the second contact pads.

2. The electronic device of claim 1, wherein the first substrate comprises a contact pad formed in an area in which the contact pin of the socket part is positioned on the lower surface of the first substrate, based on a direction in which the head part is coupled and bonded to the contact pin of the socket part, and wherein the electrical component is disposed within an opening formed in a front bracket of the electronic device, in which the opening is distinct from the through hole.

3. The electronic device of claim 1, further comprising a support member coupled to the lower surface of the first substrate, and supporting the socket part from a lower portion thereof.

4. The electronic device of claim 1, wherein the plurality of contact pads comprises: a first contact pad formed at the lower surface of the substrate, and bonded to the contact pin of the socket part; and a plurality of second contact pads coupled to the wiring of the second substrate.

5. The electronic device of claim 1, wherein the second substrate comprises socket connection wiring bonded to the first substrate and the contact pin of the socket part, and wherein the contact pin is electrically connected to the first substrate through the socket connection wiring of the second substrate.

6. The electronic device of claim 1, wherein a thickness of the second substrate is thinner than that of the first substrate.

7. The electronic device of claim 1, wherein the second substrate comprises a flexible printed circuit board (FPCB).

8. The electronic device of claim 1, further comprising an interposer substrate disposed between the first substrate and the second substrate, wherein the interposer substrate comprises a via electrically connecting the first substrate and the second substrate.

9. The electronic device of claim 1, further comprising a shielding member covering an upper surface and a side surface of the head part, and including a conductive material.

10. The electronic device of claim 3, wherein the support member includes a metal material.

11. The electronic device of claim 3, wherein the first substrate comprises an anchor affixed to the first substrate by penetration of the first substrate in the thickness direction, and wherein the anchor fixes the support member to the lower surface thereof.

12. The electronic device of claim 10, wherein the first substrate comprises ground wiring that grounds the electrical component, and wherein a coupling pad of the support member comprises a ground pad electrically connected to the ground wiring.

13. The electronic device of claim 8, wherein the interposer substrate comprises socket connection wiring bonded to the first substrate and the contact pin of the socket part, and wherein the contact pin is electrically connected to the first substrate through the socket connection wiring of the interposer substrate.

14. The electronic device of claim 9, wherein the shielding member is formed by cutting the conductive material into a shape, including a central portion and a plurality of side shielding parts extending radially from the central portion, and bending at least some of the plurality of side shielding parts to cover the side surface of the head part.

15. The electronic device of claim 14, wherein the shielding member is formed by further cutting the conductive material on a thin plate into a T-shape or a plus (+) shape.
